

Experiment 21 :

Create a Python program that performs syntax-driven semantic analysis by extracting noun phrases and their meanings from a sentence.

Code:

```
import nltk

from nltk import word_tokenize, pos_tag, RegexpParser

nltk.download('punkt')

nltk.download('averaged_perceptron_tagger')

sentence = input("Enter a sentence: ")

words = word_tokenize(sentence)

tags = pos_tag(words)

grammar = "NP: {<DT>?<JJ>*<NN.*>+}"

chunk_parser = RegexpParser(grammar)

tree = chunk_parser.parse(tags)

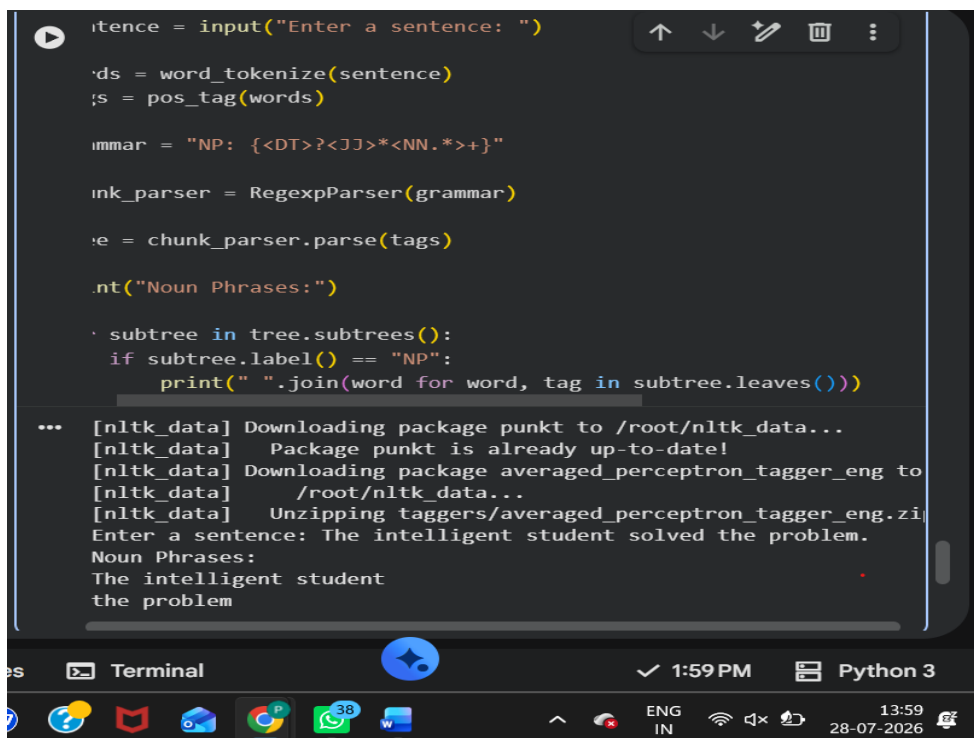
print("Noun Phrases:")

for subtree in tree.subtrees():

    if subtree.label() == "NP":

        print(" ".join(word for word, tag in subtree.leaves()))
```

Output



```
Enter a sentence: The intelligent student solved the problem.
Noun Phrases:
The intelligent student
the problem
```

The screenshot shows a terminal window with a dark background. The code from the previous block is pasted into the terminal. The output shows the program downloading the 'punkt' and 'averaged_perceptron_tagger' packages from nltk_data. It then prompts the user to enter a sentence. The user enters "The intelligent student solved the problem." The program then prints "Noun Phrases:" followed by two lines of output: "The intelligent student" and "the problem". The terminal window has a standard Linux-style taskbar at the bottom with various application icons and system status indicators.